

Craftax runs compared: achievements and what unlocks them

14 finished runs, 1.1M env steps each, size50m, one training seed per run. Scored with tools/death_eval.py on the same fixed evaluation worlds. Updated 2026-10-07.

Every number is the mean episode-final achievement count over the evaluation episodes, and each unlock rate is the share of episodes in which that achievement fired. Every run was scored on the same fixed evaluation worlds; re-scoring two old checkpoints with today's code reproduced them exactly.

The two-level agent sets the record. Manager v1.2 -- B2 plus a manager that picks a tech-tree goal and holds it until reached (or 32 steps), with one critic shared by both levels: a game-reward head judges both, a goal-reward head only the bottom actor -- reaches 9.73 achievements, against 8.53 for v1.1, 7.60 for B2 and 7.13 for v1. It is the first run to make the stone pickaxe regularly (23%), and it mines coal (10%), makes both wood tools far more often (wood sword 57%, B2 20%) and places a table in every episode. v1.1 (8.53), which pays both levels when a goal is reached, places the furnace and stone in 67% of episodes and makes a stone sword (23%) and arrows (10%); v1, whose manager never learned to choose, stays below B2. Manager v1.3 adds survival goals (drink, eat) and a survival potential to v1.2: it ties on achievements (9.67) but lives 73 steps longer (294 vs 221), dies to mobs far less (5 of 30 deaths vs 14), drinks more (80%) and reaches the dungeon for the first time (3%) -- while giving up the stone tools; hunger becomes its main killer (16 of 30).

Experiment B before it: masking impossible actions (B2) 7.60 against 5.87 for the same model without it; validity flags as an input (B1) 6.90. The non-cheating model (Honest map) matches the privileged control (5.87 vs 5.93).

Every map run except the hand-written-rewards one unlocks WAKE_UP (sleep, then wake) in 80-98% of episodes; vanilla never does.

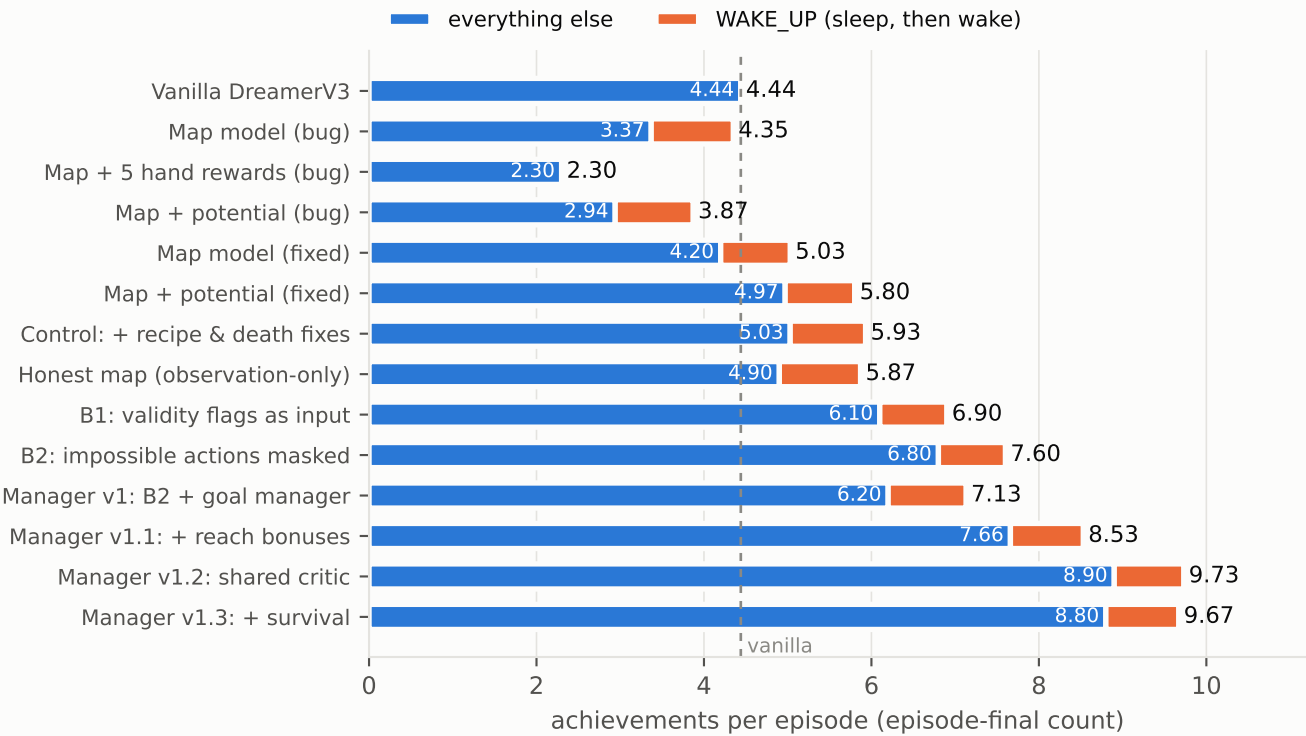
Survival is still unsolved: hunger now kills v1.3 and thirst kills v1.2. Normalized return, measured with Craftax's unshaped reward over the 30 evaluation episodes (not estimated): 3.91% for both v1.2 and v1.3 (mean return 8.83 of 226), B2 about 3.0%, against 7.09% for the best published 1M-step agent (ITC). The masking and the manager's goals use game knowledge at test time, which the published agents do not; v2, built but not yet run, replaces every piece with a learned one.

run	eps	ach.	w/o wake	life	table	pickaxe	sword	stone	drink
Vanilla DreamerV3	25	4.44	4.44	262	64%	24%	0%	12%	80%
Map model (bug)	60	4.35	3.37	294	35%	0%	0%	0%	43%
Map + 5 hand rewards (bug)	50	2.30	2.30	263	0%	0%	0%	0%	44%
Map + potential (bug)	30	3.87	2.94	277	13%	0%	0%	0%	37%
Map model (fixed)	30	5.03	4.20	264	63%	7%	0%	0%	70%
Map + potential (fixed)	30	5.80	4.97	278	90%	30%	10%	20%	60%
Control: + recipe & death fixes	30	5.93	5.03	268	83%	0%	60%	0%	67%
Honest map (observation-only)	30	5.87	4.90	243	87%	20%	20%	7%	83%
B1: validity flags as input	30	6.90	6.10	311	87%	50%	30%	33%	80%
B2: impossible actions masked	30	7.60	6.80	246	87%	53%	20%	43%	70%
Manager v1: B2 + goal manager	30	7.13	6.20	254	73%	57%	23%	47%	63%
Manager v1.1: + reach bonuses	30	8.53	7.66	259	93%	73%	33%	73%	57%
Manager v1.2: shared critic	30	9.73	8.90	221	100%	80%	57%	77%	57%
Manager v1.3: + survival	30	9.67	8.80	294	97%	83%	40%	83%	80%

eps = evaluation episodes. ach. = mean episode-final achievements; w/o wake = the same minus WAKE_UP. life = mean lifespan in steps. Rates are % of episodes. 'bug' = imag_shift True, which corrupted the policy gradient (fixed 2026-09-20). map_survive was re-evaluated on 2026-09-28 and reproduced its original 2.30. One training seed per run, so the run-to-run spread is unknown. 'Map + potential (fixed)' is the old 5.80 record; 'Control' adds the recipe and death fixes with the privileged map; 'Honest map' has the same fixes with observation-only targets. 'B1' and 'B2' are experiment B on the honest-map configuration: validity flags as an encoder input, and impossible actions masked in training. 'mgr v1' and 'mgr v1.1' are B2 plus the two-level agent (docs/design-manager.md): v1 pays the manager the game reward only; v1.1 also pays both levels when the goal is reached and masks goals that already hold; v1.2 shares one two-headed critic between both levels and holds each goal until reached or 32 steps; v1.3 adds survival goals and a survival potential to v1.2.

Sleep inflates the map runs; the manager is real progress

Mean episode-final achievements per run, split into WAKE_UP and the rest; unlock rate of the early tech-tree rungs



Unlock rate (% of episodes), early tech tree in dependency order

Vanilla DreamerV3	76	64	24	0	12	0	0	0	0
Map model (bug)	72	35	0	0	0	0	0	0	0
Map + 5 hand rewards (bug)	52	0	0	0	0	0	0	0	0
Map + potential (bug)	63	13	0	0	0	0	0	0	0
Map model (fixed)	83	63	7	0	0	0	0	0	0
Map + potential (fixed)	100	90	30	10	20	3	0	0	0
Control: + recipe & death fixes	97	83	0	60	0	0	0	0	0
Honest map (observation-only)	93	87	20	20	7	0	0	0	0
B1: validity flags as input	100	87	50	30	33	0	23	0	0
B2: impossible actions masked	93	87	53	20	43	40	40	20	0
Manager v1: B2 + goal manager	93	73	57	23	47	43	40	0	3
Manager v1.1: + reach bonuses	97	93	73	33	73	67	67	0	0
Manager v1.2: shared critic	100	100	80	57	77	73	60	10	23
Manager v1.3: + survival	100	97	83	40	83	83	83	13	0

collect wood
place table
make wood pickaxe
make wood sword
collect stone
place stone
place furnace
collect coal
make stone pickaxe

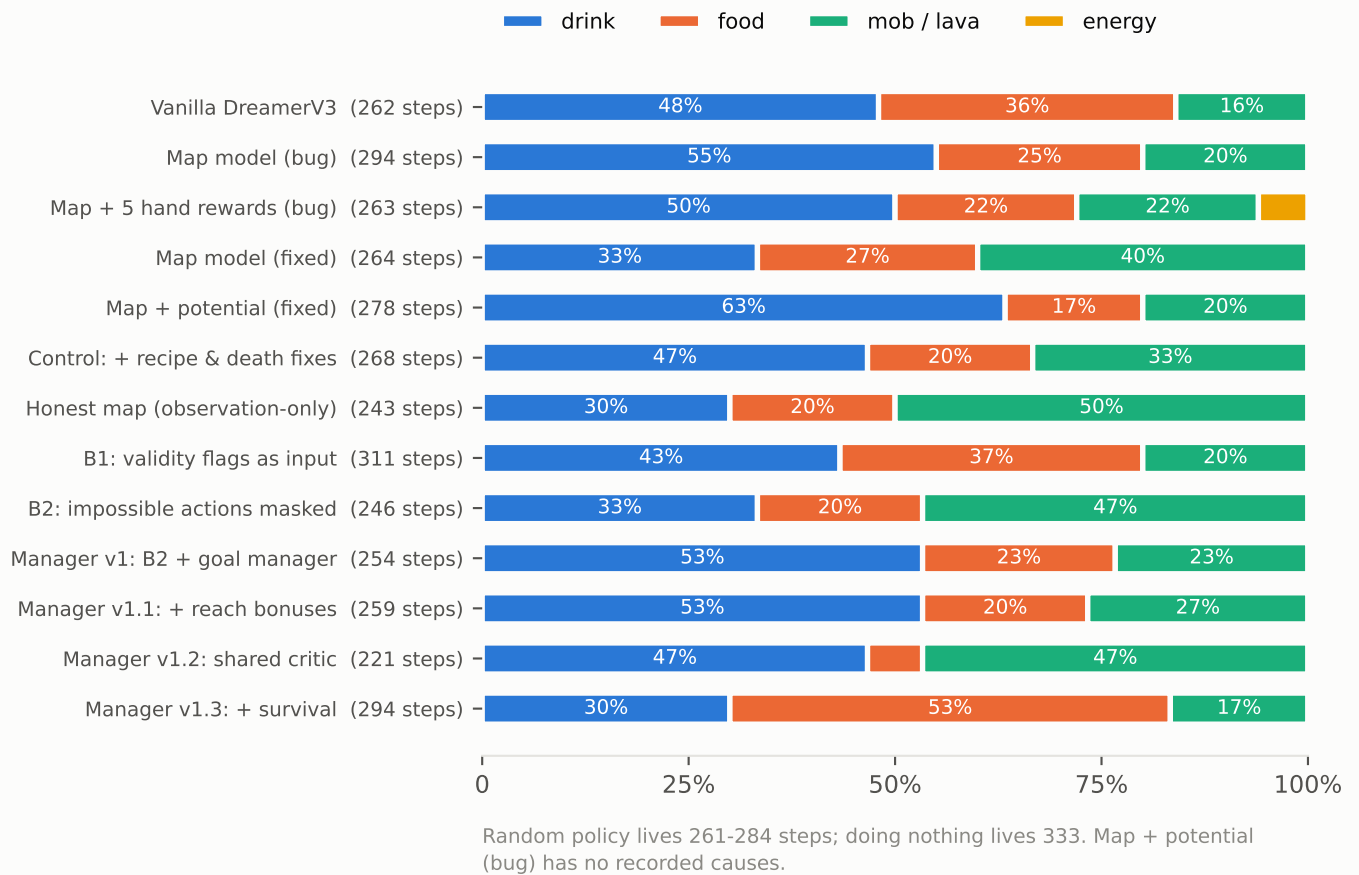
Every achievement any run unlocked

% of evaluation episodes in which each achievement fired. The other 46 of 67 were never unlocked by any run.

[illegible]

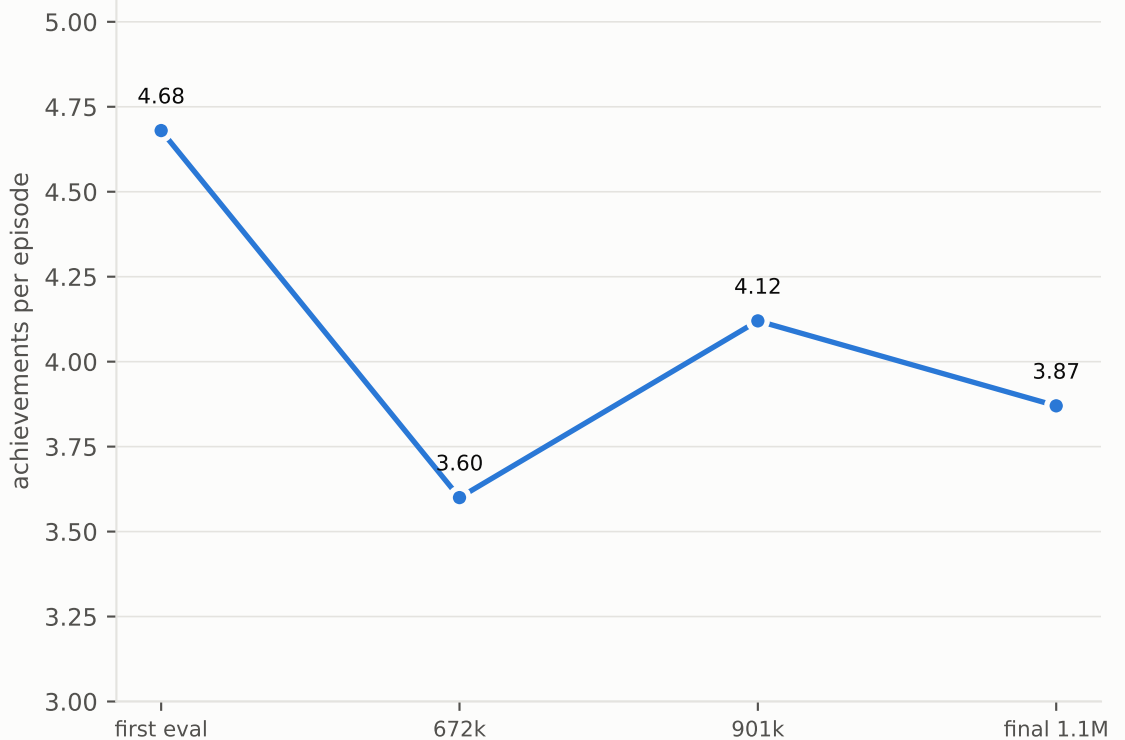
Survival never moved

Cause of death (share of evaluation episodes) and lifespan; and how much one run moves between its own checkpoints



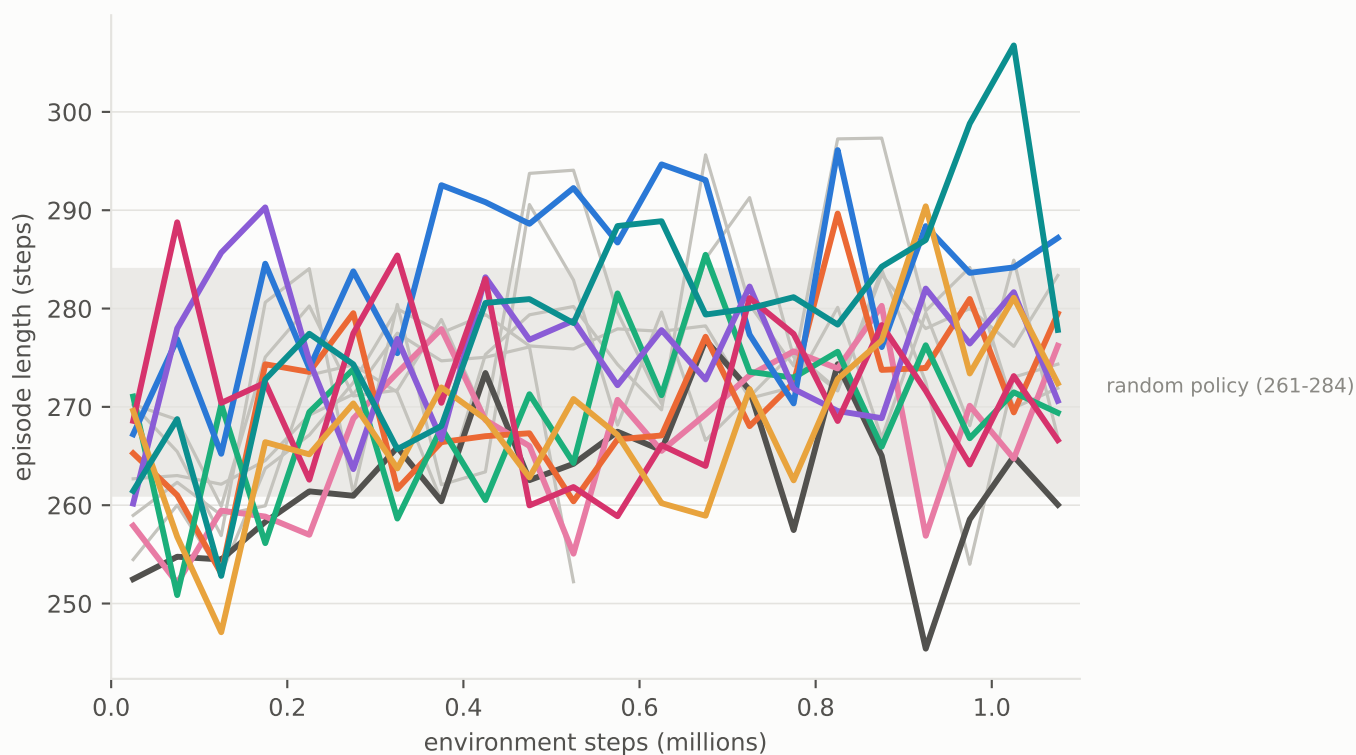
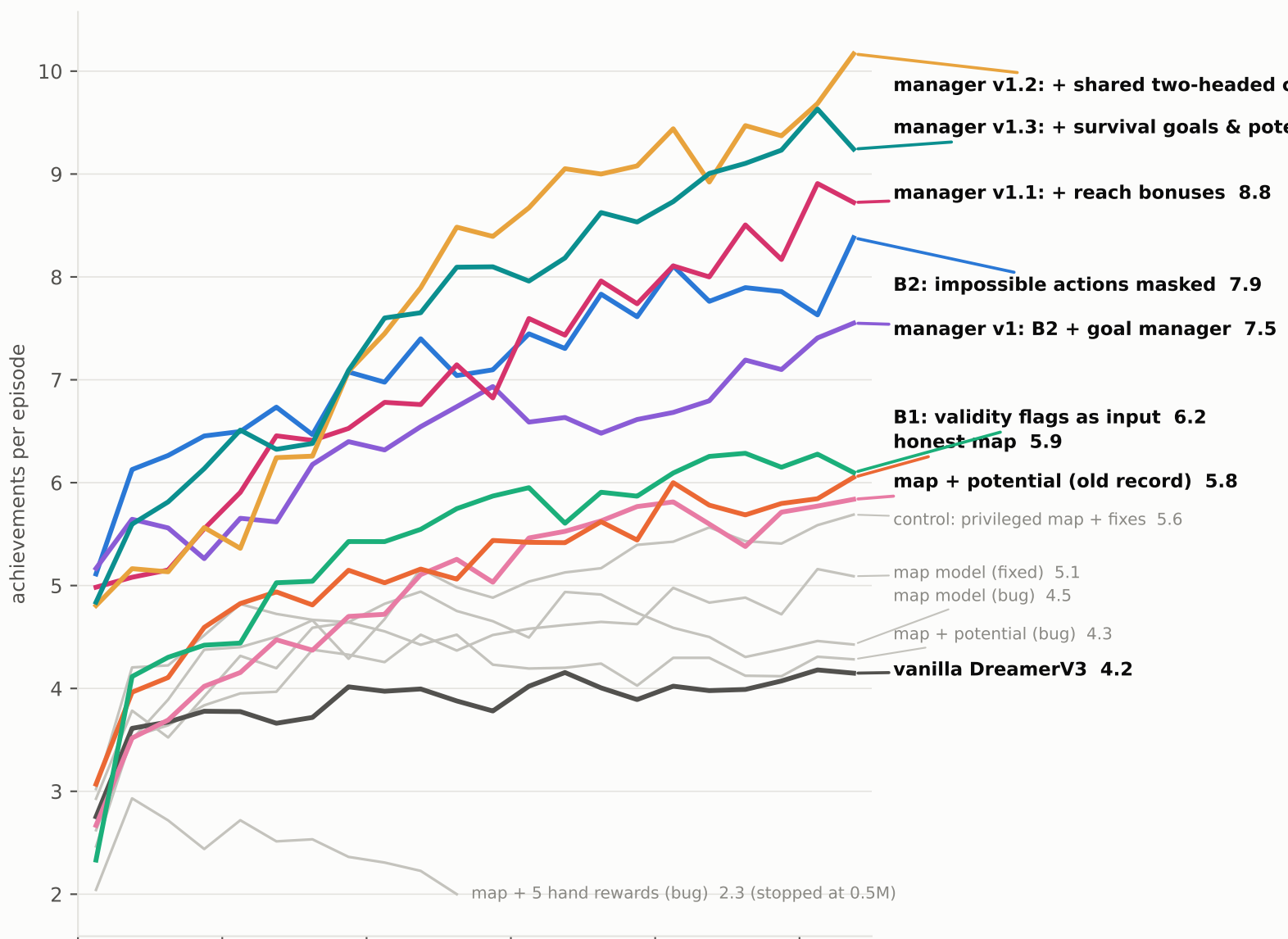
One run, four checkpoints: Map + potential (bug)

The same training run, scored at four points (25-30 episodes each), moves by 1.08 achievements. That is the scale of noise any comparison between single runs has to beat.



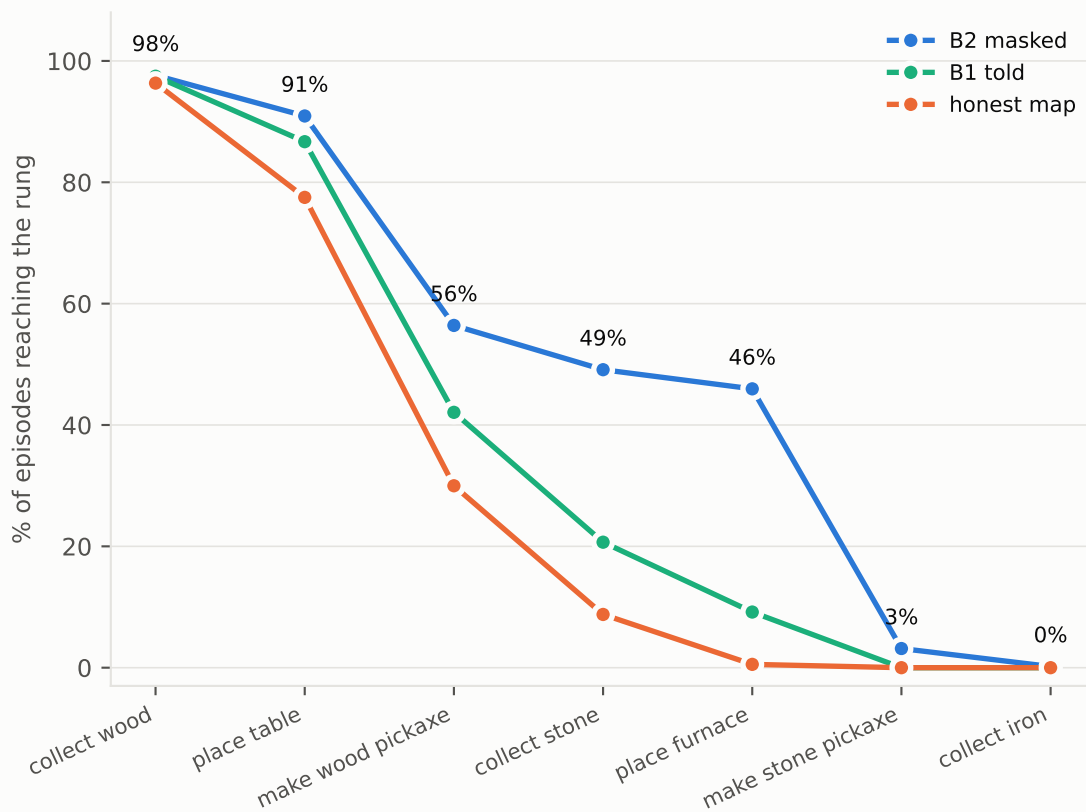
Training curves: every run

Rebuilt from each run's replay; 50k-step bins; value = mean of the last 100k steps; one seed per run.



Where B2 falls short

Last 150k training steps of each run (~500 episodes), decoded from replay by tools/diagnose_run.py



1. The stone pickaxe is the wall. B2 reaches the furnace in 46% of episodes but makes the stone pickaxe in 3%. Of the 249 episodes that collected stone, 73% held wood and stone at the same time, but only 22% ever stood next to a table while holding both, and 6% made the pickaxe. Two gaps: it rarely brings the ingredients back to a table (logistics), and when it does the chance lasts one step -- the single log usually goes into the wood sword or it walks off (measured in docs/b2-casebook.pdf).
2. Table to wood pickaxe leaks. 38% of the episodes that place a table never make the wood pickaxe. At a table holding wood it crafts every time, but every one of these episodes placed the table with its last two logs, and later logs go into another table rather than back to the first (docs/b2-casebook.pdf).
3. Survival. Training-window lifespan 285 (median 275) against 261-284 for a random policy. Thirst is still the first meter to run out in 45% of deaths and hunger in 24%: Crafttax pays for the first drink only, so topping up is never rewarded. A hostile mob is adjacent at 61% of deaths; it dies in lava in 4% (the others: under 1%), the price of exploring further. Deaths in its sleep are rare (5%).
4. Wasted presses. 43% of presses still do nothing (experiment A): DO with nothing in front (26%), NOOP and walking into walls. Masking only removes the special actions; these basic ones are never masked.

What the results say, and what would settle the rest

0. The manager pays off once it learns to choose -- most with a shared critic

v1 paid the manager the game reward only; most goals cannot be reached in 8 steps, so the actor ignored them, the manager stayed uniform, and it scored below B2 (7.13). v1.1 pays both levels when a goal is reached; its manager learned to choose and it scored 8.53. v1.2 replaces the manager's own critic with one two-headed critic shared with the bottom actor, normalises the game and goal rewards separately, and holds each goal until reached: 9.73, and the first regular stone pickaxe (23%). Its manager favours goals it cannot complete yet (iron pickaxe, iron sword, diamond), which pay the bottom actor for gathering every ingredient on the way -- plausibly why it stocks up instead of spending its wood at once. All three trailed B2 early and overtook it between 350k and 550k steps. One seed per run.

1. Masking impossible actions is the largest gain so far

B2 7.60 vs honest map 5.87, same model and reward. Experiment A on its final checkpoint shows why: buttons that sat at exactly 0% in the states where they would work now get pressed there (furnace 3.2% over 599 states, placing stone 2.4%, placing a table 75%, the wood pickaxe 100%), and every no-effect press is now a basic action (DO, NOOP, a blocked move). Once a useless press can no longer teach the agent that a button is bad, it presses that button when it works.

2. Telling helps, masking helps more

B1 (flags as input) reaches 6.90: both wood crafts survive in one run for the first time and the furnace comes off zero, but placing stone stays at 0.0% across 452 chances and about 6% of presses still go to impossible buttons. The information helps; removing the suppressing gradient helps more. Masking changes the action space relative to published Craftax baselines, so it is reported as a separate arm.

3. The next wall is sequencing, not suppression

In B2 the stone pickaxe was craftable in only 5 of 5,167 evaluation steps: the agent mines stone but rarely returns to a table holding both wood and stone. That gates iron and every 3-5-point achievement after it. It is a planning problem, so the next experiment should target multi-step sequences (experiment D, return-then-explore, or a sub-goal signal), not the action distribution.

4. Survival is still unsolved

Lifespans stay near the random band. v1.3's survival goals and potential add 73 steps over v1.2 (294 vs 221) and cut mob and lava deaths from 14 to 5 of 30, but buy no extra score: it trades the stone tools for breadth (furnace and placed stone 83%, first dungeon entries) and starves instead (hunger first in 16 of 30). Eating is the next survival target.

5. What to run next

Training now: B3 (B2 with the action mask learned instead of transcribed from the rules). Built and not yet run: v2, the two-level agent with every piece of game knowledge replaced by a learned part (learned mask, a learned goal codebook, curiosity instead of reward hints, RSSM-2 trained as a memory instead of on map labels) -- the version comparable with the published agents. Second seeds of the leaders are still missing; every comparison here is single-seed.